Nhat-Minh Le-Phan

Born: 05 Sep 1999, Nghean, Vietnam
PhD student, School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore
phannhat001@e.ntu.edu.sg — (+84) 0889831099 — www.linkedin.com/in/Nhat-Minh Le-Phan — Google Scholar: Nhat-Minh Le-Phan

RESEARCH INTERESTS

Multi-agent Control, Intelligent Control, Resilient Control, Nonlinear Control, Missile Guidance and Control , . . .

EDUCATION

Nanyang Technological University, Singapore — August 2025 — Present
PhD Student in Control Theory, School of Electrical and Electronic Engineering — Cumulative GPA: 5.00/5.00
Supervisor: Prof. Lihua Xie

Hanoi University of Science and Technology, Hanoi, Vietnam — July 2022 — Oct 2024
Master of Science in Control and Automation Engineering — Cumulative GPA: 3.95/4.00
Thesis Title: Randomized Matrix Weighted Consensus and Its Applications — Valedictorian
Thesis Grade: 10/10
Supervisor: Prof. Phuoc Doan Nguyen

Hanoi University of Science and Technology, Hanoi, Vietnam — August 2017 — May 2022
Bachelor of Science: Control and Automation Engineering (5-years program) — Cumulative GPA: 3.88/4.00
Ranked 1st in Vietnam in Electrical and Electronic Engineering by QS Rankings. — Valedictorian, rank: 01/496
Supervisor: Assoc. Prof. Nga Thi-Thuy Vu

Phan Boi Chau High School for the Gifted, Nghean, Vietnam — August 2014 — July 2017
Major in Chemistry

EXPERIENCES

Hanoi University of Science and Technology (HUST) — Hanoi, Vietnam
Faculty at Department of Automation Engineering — August 2024 — Present

- Member of Multi-agent control system laboratory
- Teaching assistant for undergraduate-level courses

Viettel Aerospace Institute (VTX) — Hanoi, Vietnam
Guidance and Control Engineer — July 2022 — August 2024

- Member of Control technologies department
- Organize lectures on advanced control theory for flight vehicles
- Design and develop advanced nonlinear/robust/adaptive/intelligent controllers for flight vehicles
- Design and develop advanced guidance algorithms for flight vehicles

Research and Development in Control Systems Laboratory, HUST — Hanoi, Vietnam
Student Member — 2019 — 2022

- Design nonlinear control algorithms for Wind Energy Control Systems
- Design nonlinear control algorithms for Robotic Control Systems

PROJECTS

Design a reinforcement learning based adaptive tuning law for the Three-loop Autopilot controller — VTX (signed an NDA)
Guidance and Control Engineer — December 2023 — August 2024

- Apply the DDPG algorithm to the framework of Three-loop Autopilot controller
- Design the training scenario and hyperparameter tuning
- Evaluate the performance of the proposed algorithm with the traditional gain-scheduling method using the Monte Carlo technique.

Design adaptive controllers for the roll stabilization problem of flight vehicles — VTX (signed an NDA)
Guidance and Control Engineer — July 2022 — July 2023

- Research on flight dynamics and modeling methods for flying vehicles
- Design an adaptive controller for the roll stabilization problem of flight vehicles.
- Evaluate the performance of the proposed algorithm with traditional controllers using the Monte Carlo technique.

PUBLICATIONS

Journal Papers

Published

- **Nhat-Minh Le-Phan**, Thien-Minh Nguyen, Wenjie Liu, and Lihua Xie*, "Resilient Tracking of Bearing-Constrained Leader–Follower Formations Under Denial-of-Service Attacks," in IEEE Transactions on Control of Network Systems, 2026, doi:10.1109/TCNS.2026.3719506.
- Tat-Thang Nguyen, **Nhat-Minh Le-Phan** and Nga Thi-Thuy Vu*, "Data-Driven Q-Learning Based Robust Longitudinal Missile Autopilot Design Without Angle-of-Attack Measurement", International Journal of Aeronautical and Space Sciences, 2026.
- Duong Tuan Nguyen, **Nhat-Minh Le-Phan** and Nga Thi-Thuy Vu*, "Performance Improvement of Proton Exchange Membrane Fuel Cells via Deep Q-Network", Energy Systems, 2026, doi: 10.1007/s12667-026-00787-2.
- **Nhat-Minh Le-Phan**, Phuoc Doan Nguyen, Hyo-Sung Ahn, Minh Hoang Trinh*, "Bearing-Only Solution for Fermat-Weber Location Problem: Generalized Algorithms", Automatica, Volume 176, 2025, 112242, ISSN 0005-1098, doi: 10.1016/j.automatica.2025.112242.
- Minh Hoang Trinh*, Hoang Huy Vu, **Nhat-Minh Le-Phan**, Quyen Ngoc Nguyen, "Matrix-Scaled Consensus over Undirected Networks," in IEEE Transactions on Network Science and Engineering, doi: 10.1109/TNSE.2024.3471710.
- **Nhat-Minh Le-Phan**, Minh Hoang Trinh*, and P. D. Nguyen, "Randomized Matrix Weighted Consensus," in IEEE Transactions on Network Science and Engineering, vol. 11, no. 4, pp. 3536-3549, July-Aug. 2024, doi: 10.1109/TNSE.2024.3376643.
- **Nhat-Minh Le-Phan***, D. L. Dinh, A. N. D. Duc and N. P. Van, "Inverse Optimal-Based Attitude Control for Fixed-Wing Unmanned Aerial Vehicles," in IEEE Access, vol. 11, pp. 52996-53005, 2023, doi: 10.1109/ACCESS.2023.3280424.
- **Nhat-Minh Le-Phan**, Duong-Nguyen Tung, Tung-Nguyen Thanh, Nga Thi-Thuy Vu* "ANFIS wind speed estimator based output feedback near-optimal MPPT control for PMSG wind turbine", Journal of Control, Automation, and Electrical System, doi: 10.1007/s40313-022-00980-5

Submitted/Preprints

- Minh Hoang Trinh, **Nhat-Minh Le-Phan** and Hyo-Sung Ahn, "The networked input-output economic problem", arxiv, doi: 2412.13564
- Cuong M. Nguyen, **Nhat-Minh Le-Phan** and Nam H. Nguyen, "Robust UAV formation control with collision avoidance via predefined-time attitude control and state observer"

Conference Papers

Published

- Tien Dat Vu, Phuoc Vinh Nguyen, **Nhat-Minh Le-Phan**, and Minh N. Doan, "Complete Controllability of Matrix-Weighted Leader–Follower Networks," 2026 65th IEEE Conference on Decision and Control (CDC).
- Quang-Dai Pham, **Nhat-Minh Le-Phan**, Chenhao Zhao, and Christopher H. T. Lee , "Super-Twisting Sliding Mode Disturbance Observer-Based Neural-Network Controller for Motor Drives," 2026 International Conference on Electrical Machines and Systems (ICEMS2026).
- **Nhat-Minh Le-Phan**, Minh Hoang Trinh*, Phuoc Doan Nguyen, "Dimensional lifting, finite time convergence, and bearing-only solutions to the Fermat-Weber Location Problem," Proc. of the 12th International Conference on Control, Automation and Information Sciences (ICCAIS), Hanoi, Vietnam, 2023.
- **Nhat-Minh Le-Phan**, Minh Hoang Trinh*, Phuoc Doan Nguyen, "Bearing-Based Network Localization Under Gossip Protocol," **Oral presentation** at 12th International Conference on Control, Automation and Information Sciences (ICCAIS), Hanoi, Vietnam, 2023.

Domestic Journal/Conference Papers

Published

- Dung Van Vu, Hieu Minh Vu, **Nhat-Minh Le-Phan**, Hieu Minh Nguyen, Tuynh Van Pham, Minh Hoang Trinh*, "Control of double-integrator systems without velocity measurements and application of matrix-scaled consensus problem," Proc. of the 7th Vietnam International Conference and Exhibition on Control and Automation (VCCA), Hai Phong, Vietnam.
- **Nhat-Minh Le-Phan**, Tung Thanh Nguyen, Duong Tung Nguyen, Nga Thi-Thuy Vu*, "Output Feedback Adaptive Pitch Angle Control for Variable Speed Wind Turbine, "Journal of Science and Technology: Smart Systems and Devices, 2023.

ACADEMIC SERVICE

Technical reviewer
Journals: Automatica, IEEE Transactions on Network Science and Engineering, IEEE Control System Letter, Unmanned Systems, Science China Information Sciences.
Conference: American Control Conference (ACC), Conference on Decision and Control (CDC).

SELECTED COURSES

Graduate Level Courses

- Control of Multi-agent systems
- Linear System
- Analysis and Control of Nonlinear Systems
- Reinforcement Learning Based Control
- Optimal Control
- Advanced Process Control
- Genetic Algorithm and Machine Learning
- Convex Optimization
- Robotic and Intelligent Sensors
- Neural Networks and Deep Learning

Bachelor's Courses

- Linear Algebra
- Calculus
- Probability and Statistic
- Linear Control Theory
- Nonlinear Control
- Networked Control
- Control of Mechatronic Systems
- Process Control
- Fundamental of Neural Network and Fuzzy System

SCHOLARSHIPS/AWARDS

NTU PhD Research Scholarship — NTU, Singapore
Full tuition fee waiver and a monthly stipend. — 2025-2029

Viettel excellence scholarship — HUST, Hanoi, Vietnam
Full Master's Scholarship awarded in recognition of outstanding academic achievement as a master's student. — 2024

Student Scientific Research — HUST, Hanoi, Vietnam
Received the third prize for the topic: Application of near-infrared spectral and machine learning methods to sugar quality management — 2022

Academic Incentive Scholarship — HUST, Hanoi, Vietnam
Tuition support for undergraduate students with excellent academic performance. — 2020,2021

REFERENCES

Lihua Xie
Fellow of IEEE, Fellow of IFAC
Full Professor, President's Chair in Control Engineering, Nanyang Technological University, Singapore
E-mail: ELHXIE@ntu.edu.sg
Scholar Profiles: Personal Page — Google Scholar

Minh Hoang Trinh, PhD
Lecturer, East Asia University of Technology, Hanoi, Vietnam
E-mail: minhtrinh@ieee.org
Scholar Profiles: Personal Page — Google Scholar — LinkedIn

Phuoc Doan Nguyen
Full Professor
Department of Automation, Hanoi University of Science and Technology, Hanoi, Vietnam
Member of The State Council for Professor of Vietnam
E-mail: phuoc.nguyendoan@hust.edu.vn
Scholar Profiles: Google Scholar